

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2461194.324 +/- 0.0019 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.204 +/- 0.016
Transit depth (Rp/Rs)^2	4.17 +/- 0.66 [%]
Orbital Inclination (inc)	89.84 +/- 0.63
Airmass coefficient 1 (a1)	1.498 +/- 0.014
Airmass coefficient 2 (a2)	-0.245 +/- 0.062
Scatter in the residuals of the lightcurve fit is	0.93 %
Best Comparison Star	None
Optimal Aperture	2.82
Optimal Annulus	10.15
Transit Duration (day)	0.0208 +/- 0.0021

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.204 ± 0.016 vs 0.14762 (deviation 3.52σ)
Duration fit vs published	0.0208 d vs 0.1033 d (deviation 39.3σ)
Mid-time O–C	-5007.5 min
Depth SNR	12.7
Residual scatter	0.93 %

Light curve

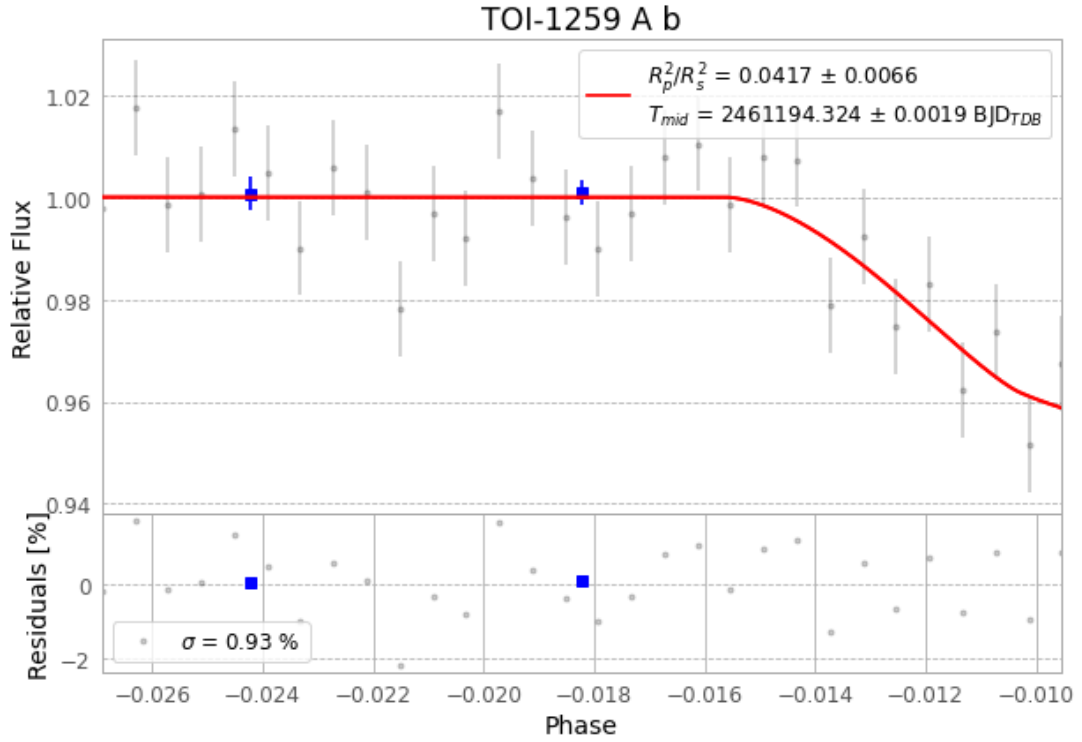


Figure 1 — FinalLightCurve_TOI-1259 A b_2026-06-05.png (SHA-256 a65143334545fd47...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [299, 292], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 dabaec1ffaf85635...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

30 FITS frames (20 MB), TOI1259260606050015.FITS → TOI1259260606062715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TOI1259-260606.log (2c52a2883c9e...), FinalLightCurve_TOI-1259 A b_2026-06-05.png (a65143334545...), FinalLightCurve_TOI-1259 A b_2026-06-05.csv (b59646689f51...)

Compute resources

Wall time 82.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-16 00:04Z.